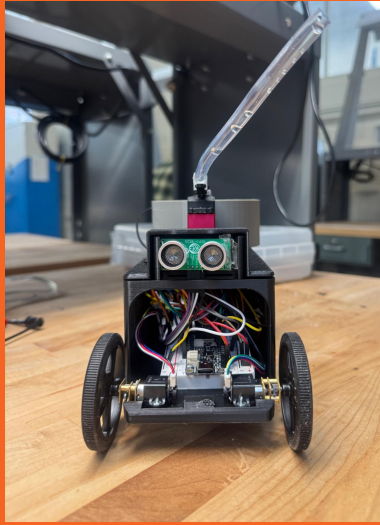




ME100 Project Presentation: RC Fire Truck



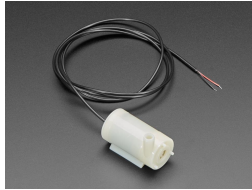
Tamira Shany, Nadia Liu, Aiden Man

Overview

This vehicle assists with fire fighting while minimizing human exposure to extreme heat and structural dangers. The rover features a dual-mode system: a remote-operated driving and turret-aiming system, and an autonomous safety override that prevents collisions.



Sensing, Actuation, and Data Transmission



Sensing:

- Ultrasonic Sensor: Distance and Obstacles
- IMU: Acceleration and Internal Temperature

Actuation:

- Water Pump
- High Torque Servo Motor
- DC Drive Motors

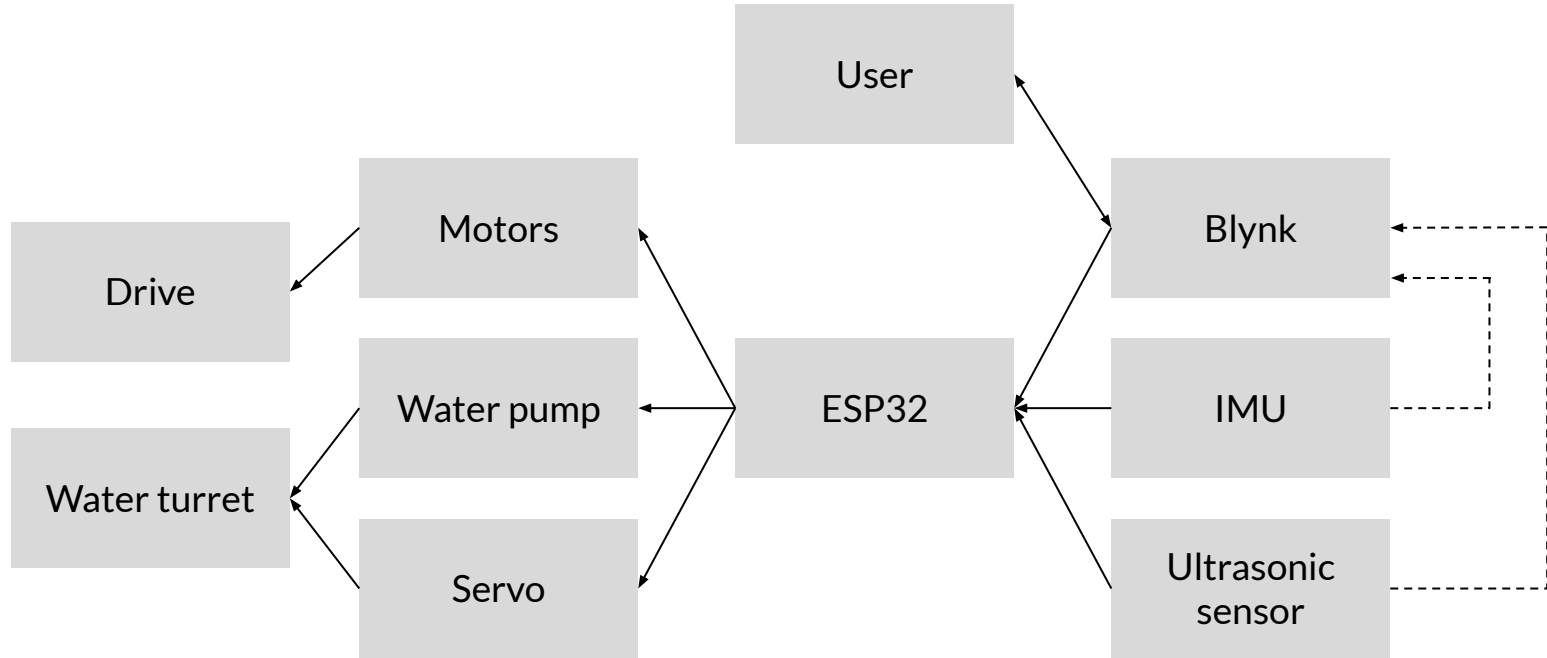
Data transmission:

- Interface: Blynk
 - Joystick for Car Control
 - Toggle for Water Pump
 - Slider for Servo Turret
 - Ultrasonic Sensor reading for distance
 - IMU reading for 3-axis acceleration

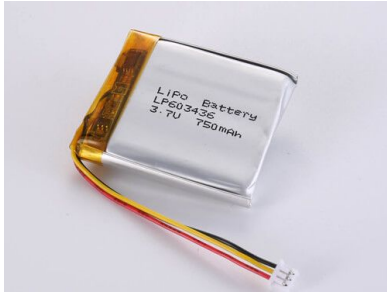


B

Sensing, Actuation, and Data Transmission



Power Distribution



Servomotor - Li-Po 4.15V

Water Pump - Li-Po 4.15V

Ultrasonic Sensor - ESP32 3.3V

IMU - ESP32 3.3V

Drive Motors - USB 5V



Functionality

Proximity detection using the Ultrasonic Sensor

- Applies safety brake to the drive motors when distance sensed < 10 cm to prevents collisions

Servo-actuated Turret

- Provides aiming for water pump with 180 degree range of motion

Water Pump

- Pressurizes the water to shoot at fires
-

Details

Servo:

- Uses Micropython-servo module, enabling direct PWM signal for positioning from simple python inputs

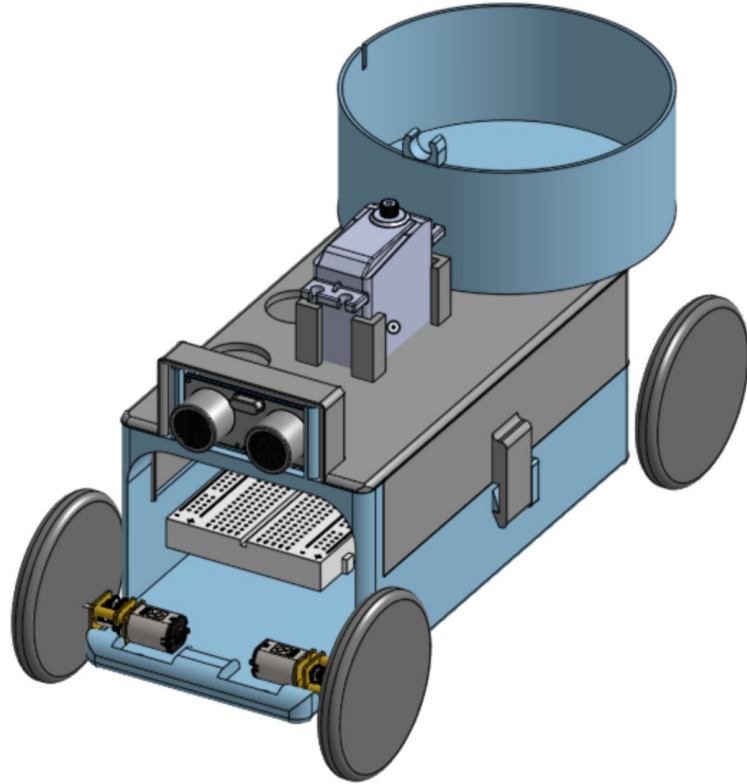
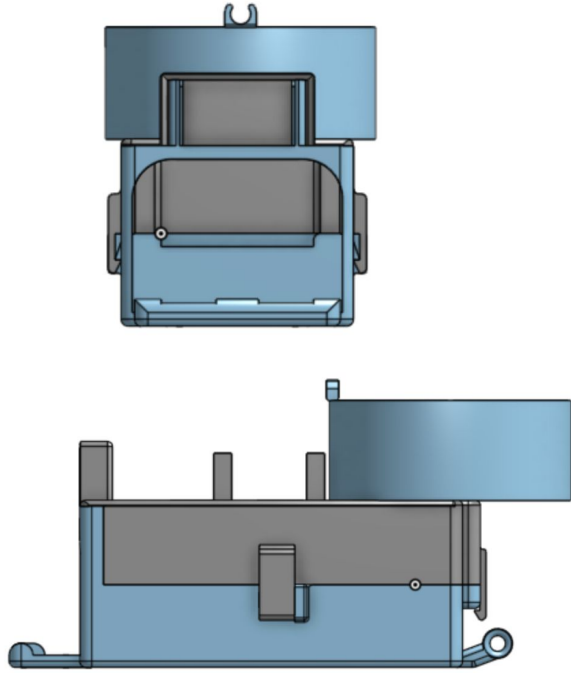
Water Pump:

- Uses NMOS gate to enable PWM signal from the ESP32 to unidirectionally actuate DC motor

Drive Motors:

- Uses DRV8833 H-Bridge for dual PWM drive
-

Physical Design



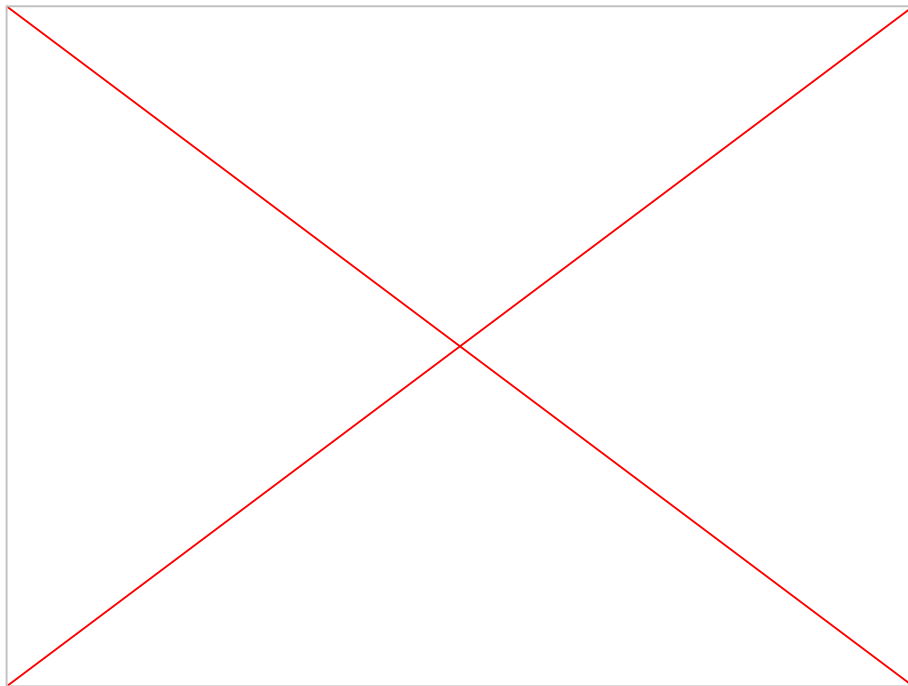
Challenges and Solutions

1. **Power Routing** - We had issues with using 4 actuators and routing power between these devices. We began trying to power everything with the USB power, but soon realized we would need external power sources. We are now using 2 additional Li-Po batteries connected to the actuators.
 2. **Water pump/Servo EMI** - While turning on the water pump, since both of the motors are inductive, the water pump creates an electromagnetic interference (EMI) with the servo. To fix this, we separated the servo wiring from the water pump wiring, created a specific ground for all the motor wires at “one point”, and added suppression capacitors as well as a flyback diode to ensure there was reduced interference as the motors turn.
-

Challenges and Solutions

1. **Wiring issues (one ESP32)** - Since we are using 1 ESP32, there is limited space on the bread board as well as with the ESP32 pins. We have changed pins a couple of times between analog and digital depending on the type that we needed.
 2. **Blynk code** - Connecting and interfacing with the Blynk app had its own learning curve. We initialized the blynk protocols and in the beginning had trouble connecting to wifi which would allow us to use Blynk through our phones. On the Blynk interface, we have 10 Datastreams, which is a lot of data sending every second. We continued to tweak the frequencies and internal timers to connect to wifi.
-

Video & Live Demo!



The End

Thank you for your time!

We had a lot of fun with ideating during the project and were able to troubleshoot all of our issues. It shows our growth as engineers, and provides us knowledge on IoT.

